Heights & Distances

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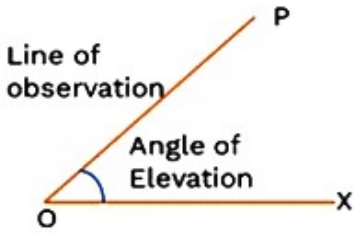
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Heights & Distances

ANGLES OF ELEVATION AND DEPRESSION

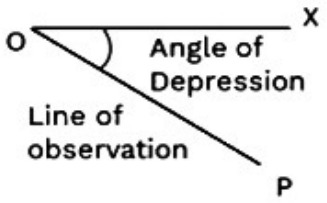
(i) Angle of elevation:

Let P be an object and OX be a horizontal line; an observer at O to perceive the object P has to elevate his eye from the direction OX to the direction OP. As such, we define $\angle XOP$ as the **angle of elevation** of P at O.



(ii) Angle of depression:

Let P be an object and OX be a horizontal line; an observer at O to perceive the object P has to depress his eye from the direction OX to the direction OP. As such, we define $\angle XOP$ as the angle of depression of P at O.

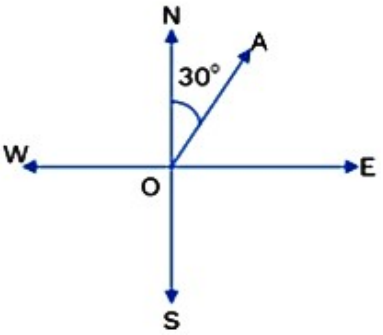


(iii) Angular elevation:

If PM is perpendicular to OX, then the angle XOP is also called the altitude or the **angular elevation** of MP at O.

Let NS and EW stand for lines in the north-south and east-west directions respectively then the acute angle which the line OA makes with NS is called the bearing of the point A from O.

The bearing of A may be indicated precisely, by giving the size of the angle and specifying whether it is measured from ON (or OS) and whether to east (or west). For example, in figure, OA is in the direction 30° east of north. Thus, the bearing is written as N 30° E.

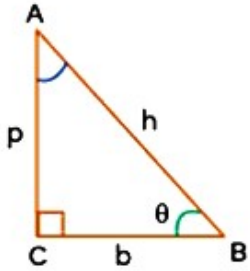


To express one side of a right angled triangle in terms of other sides :

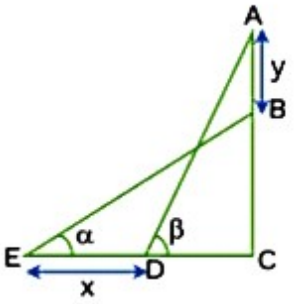
Let AB = h (hypotenuse) and let $\angle ABC = \theta$. The side opposite to θ is a perpendicular p and the remaining side is base b.

$b = p(\cot\theta)$ $\quad$ $p = b(\tan\theta)$

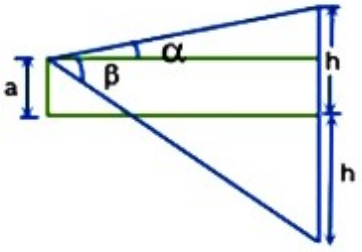
$p = h(\sin\theta)$ $\quad$ $b = h(\cos\theta)$



(a) If AD = BE, then $x = y\tan\left(\frac{\alpha+\beta}{2}\right)$



(b) $h = \frac{a\sin(\alpha+\beta)}{\sin(\beta-\alpha)}$



Example :

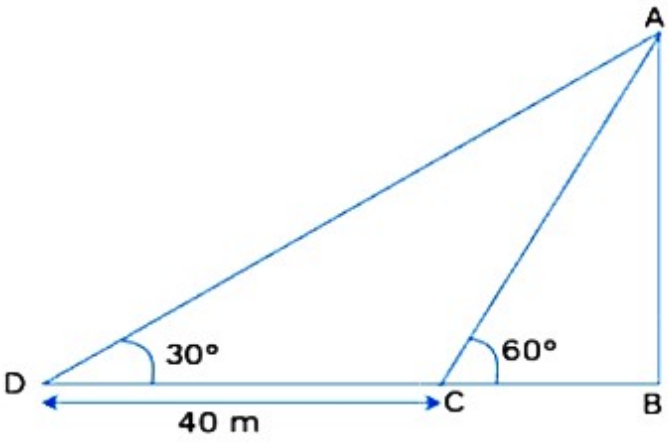
Q. **When the Sun's altitude is 30° then the shadow of a tower standing on a level ground is 40 m longer than the shadow present when Sun's altitude is 60°. Find the height of the tower.**

Sol. In figure,

let length of tower = AB and length of tower's shadow = BC when the Sun's altitude is 60°.

Hence, the top of the tower has the angle of elevation from the tip of the shadow equal to 60°.

When the angle of elevation is 30°, the length of the shadow is BD.



Now, let BC be x m and AB be h m. As, length of BD is 40 m than that of BC
Therefore BD is equal to 40 + x m
Now

In ΔABC, $\tan 60^\circ = \frac{AB}{BC}$

or $\sqrt{3} = \frac{h}{x}$ ...(1)

In ΔABD $\tan 30^\circ = \frac{AB}{BD}$

i.e., $\frac{1}{\sqrt{3}} = \frac{h}{x+40}$ ...(2)

From (1), we have $h = x\sqrt{3}$

From (1) and (2) $(x\sqrt{3})\sqrt{3} = x + 40$

$\therefore \quad 2x = 40$

So, $h = 20\sqrt{3}$ [From (1)]

$\therefore$ height of tower = $20\sqrt{3}$m.

Q. **The angles of depression of the foot and the top of an 8 m tall tower from the top of a multi-storeyed building are 45° and 30°, respectively. Find the distance between the tower and the building as well as the height of the multi-storeyed building.**

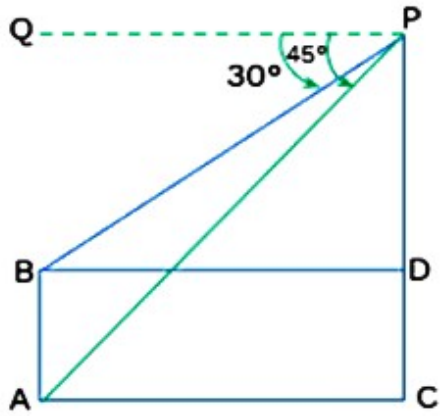
Sol. In the given fiure

PC = multi storeged building

AB = 8m tall tower

To find PC and AC

Observe that BD || PQ || AC and AP is transersal line



$\Rightarrow \angle QPA = \angle PAC = 45°$ [$\because$ alternate angle]

$\Rightarrow \angle QPB = \angle PBD = 30°$ [$\because$ alternate angle]

In right ΔPBD, we have

$$\frac{PD}{BD} = \tan 30^\circ = \frac{1}{\sqrt{3}} \text{ or } BD = PD\sqrt{3}$$

In right ΔPAC, we have

$$\frac{PC}{AC} = \tan 45^\circ = 1$$

$\Rightarrow$ PC = AC

Also, PC = CD + PD, therefore, CD + PD = AC.

Since, BD = AC and AB = CD = 8 m, we get PD + 8 = BD = $PD\sqrt{3}$

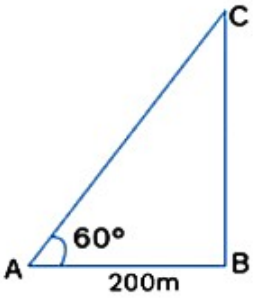
This gives $PD = \frac{8}{\sqrt{3}-1} = \frac{8(\sqrt{3}+1)}{(\sqrt{3}+1)(\sqrt{3}-1)} = 4(\sqrt{3}+1)$m

Hence, the height of the multi-storeyed building is $\{4(\sqrt{3}+1)+8\}$m = $4(3+\sqrt{3})$m

So distance between the buildings = $4(3+\sqrt{3})$m

Q. **Ray from eye of the observer to the top of tower makes angle 60° at 200 meter away from foot of tower. Find height of tower.**

Sol. In figure



Height of tower = BC

and AB = 200 m

In ΔABC, $\tan\theta = \frac{BC}{AB}$

$\therefore \quad \tan 60^\circ = \frac{BC}{200}$

$\Rightarrow \sqrt{3} = \frac{BC}{200} \Rightarrow BC = 200\sqrt{3}\,m$

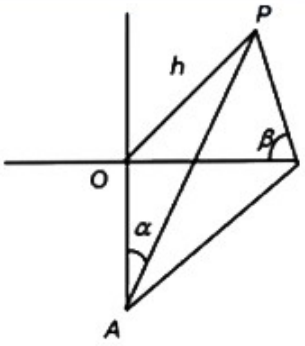
Hence, height of the tower is $200\sqrt{3}\,m$.

Q. **The angle of elevation of top of building from point A on ground due south of building is α and from point B on ground due East of building is β. If AB = d, then building's height will be**

(A) $\frac{d}{\sqrt{\tan^2\alpha - \tan^2\beta}}$ **(B)** $\frac{d}{\sqrt{\tan^2\alpha + \tan^2\beta}}$

(C) $\frac{d}{\sqrt{\cot^2\alpha + \cot^2\beta}}$ **(D)** $\frac{d}{\sqrt{\cot^2\alpha - \cot^2\beta}}$

Sol. **(C)**



$OB = h\cot\beta, OA = h\cot\alpha$

$h^2 = \frac{d^2}{\cot^2\beta + \cot^2\alpha}$

$\Rightarrow \Delta = \left(\frac{\sqrt{3}a^2}{4}\right) = 3\sqrt{3}\,sq.cm, \quad \therefore R = \frac{abc}{4\Delta} = 2cm.$

Q. **Observer standing on top of the 80 m high tower observers angle of depression of the top of the pole be 45° and that of bottom of pole is 60°. Find diff. between height of pole and tower.**

Sol. Let height of tower = MN & height of pole = LP = h

We can observe

$\angle NPQ = 45°$ & $\angle NLM = 60°$

Let PQ = x = LM

In ΔPQN

$$\tan 45° = \frac{NQ}{PQ}$$

$$\Rightarrow 1 = \frac{80 - h}{x} \Rightarrow x = 80 - h \quad \ldots(i)$$

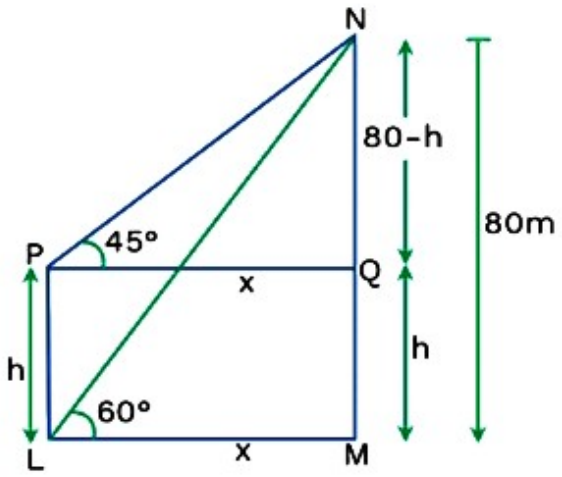
In ΔLMN

$$\tan 60° = \frac{NM}{LM} \qquad \sqrt{3} = \frac{80}{x}$$

$$x = \frac{80}{\sqrt{3}} \quad \ldots(ii)$$

$\therefore$ Difference between height of tower & pole is

$$80 - h = x = \frac{80}{\sqrt{3}} \quad \text{(from (i) \& (ii)).}$$



Q. **A vertical tower consists of two parts, the upper part being two third of the whole. At a point on the horizontal plane through the base of the tower and at a distance of 20 meters from it, the upper part of the tower subtends an angle whose tangent is $\frac{1}{2}$. The possible heights of the tower are**

(A) 20 m and $20\sqrt{3}$ m **(B) 20 m and 60 m**

(C) 16 m and 48 m **(D) None of these**

Sol. **(B)**

$\frac{H}{3}\cot\alpha = d$

$\frac{H}{3d} = \tan\alpha$ and $\frac{H}{d} = \tan\beta$

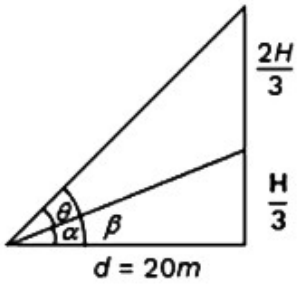
$\tan(\beta - \alpha) = \frac{1}{2} = \frac{\frac{H}{d} - \frac{H}{3d}}{1 + \frac{H^2}{3d^2}}$

$\Rightarrow 1 + \frac{H^2}{3d^2} = \frac{4H}{3d}$

$\Rightarrow H^2 - 4dH + 3d^2 = 0$

$\Rightarrow H^2 - 80H + 3(400) = 0$

$\Rightarrow H = 20$ or 60 m.



Q. **Two cars on either side of tower of unknown height are being observe. At the moment from top of tower cars are making angle of depression equal to 60° & 45° and distance between them at that time is $200\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$m, then find the height of the tower.**

Sol. Let A & B are two cars on either side of the tower CD. We can find.

$\angle CBD = 45°$ and $\angle CAD = 60°$

Given, $AB = \frac{200(\sqrt{3}+1)}{\sqrt{3}}$m

In ΔBDC, $\angle D = 90°$

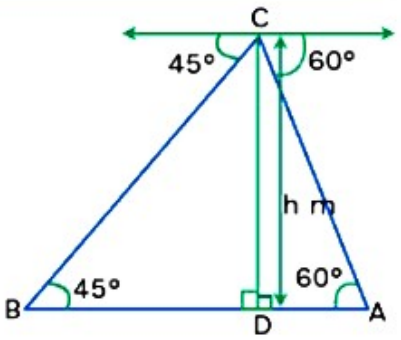
$\therefore \frac{CD}{BD} = \tan 45°$

$\Rightarrow \frac{CD}{BD} = 1 \Rightarrow CD = BD$ ...(i)

Again, in ΔADC, $\angle D = 90°$

$\therefore \frac{CD}{AD} = \tan 60° \Rightarrow \frac{CD}{AD} = \sqrt{3} \Rightarrow \frac{CD}{\sqrt{3}} = AD$ ...(ii)

(i) + (ii) give



$$AD + BD = \frac{CD}{\sqrt{3}} + CD \Rightarrow AB = CD\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$$

$$\Rightarrow 200\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right) = CD\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right) \qquad \left(\because AB = \frac{200(\sqrt{3}+1)}{\sqrt{3}}m\right)$$

On comparing, we get
CD = 200
Hence, the height of tower is 200 m.

Q. **An observer from some height observes angle of depression of car moving toward him at different times. At time t1 angle of depression be 30°. After driving for three minutes angle of depression increases by 30°. Find the time that car will take to reach the free.**
(A) 4 min. **(B) 4.5 min.** **(C) 1.5 min.** **(D) 2 min.**

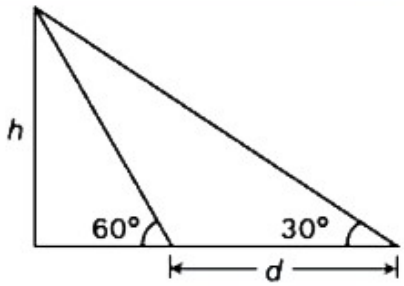
Sol. **(C)**

$d = h\cot 30° - h\cot 60°$ and time = 3 min.

$\therefore$ Speed $= \frac{h(\cot 30° - \cot 60°)}{3}$ per minute

Total distance travelled $= h\cot 60°$ in

$\frac{h\cot 60° \times 3}{h(\cot 30° - \cot 60°)}$ = 1.5 minute.



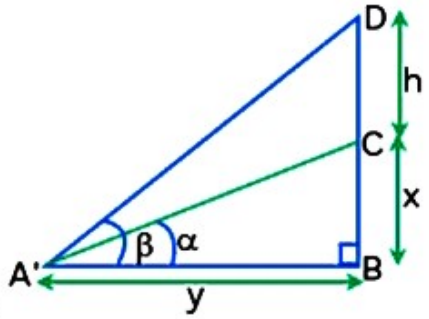
Q. **A building is standing on a horizontal plane. A vertical flagstaff of height h is mounted on top of it. From a point on the plane, the angle of elevation of the top and bottom of the flagstaff is β and α. Show that the height of the building is** $\frac{h\tan\alpha}{\tan\beta - \tan\alpha}$

Sol. Let the height of the building BC be x and CD be the flagstaff. Let CD = h.

Let A be the given point on the plane.
In ΔABC, right angled at B, we have

$\tan\alpha = \frac{BC}{AB}$

$\Rightarrow \tan\alpha = \frac{x}{y} \Rightarrow y = \frac{x}{\tan\alpha}$ ...(1)



In ΔABD, right angled at B, we have

$$\tan\beta = \frac{BD}{AB} \Rightarrow \tan\beta = \frac{h+x}{y} \Rightarrow y = \frac{h+x}{\tan\beta} \qquad ...(2)$$

From (1) and (2), we get

$$\frac{x}{\tan\alpha} = \frac{h+x}{\tan\beta} \Rightarrow \frac{x}{\tan\alpha} = \frac{h}{\tan\beta} + \frac{x}{\tan\beta}$$

$$\Rightarrow \frac{h}{\tan\beta} = \frac{x}{\tan\alpha} - \frac{x}{\tan\beta}$$

$$\Rightarrow \frac{h}{\tan\beta} = x\left(\frac{1}{\tan\alpha} - \frac{1}{\tan\beta}\right)$$

$$\Rightarrow \frac{h}{\tan\beta} = \frac{x(\tan\beta - \tan\alpha)}{\tan\alpha\tan\beta}$$

$$\Rightarrow x = \frac{h\tan\alpha \times \tan\beta}{\tan\beta(\tan\beta - \tan\alpha)} \Rightarrow x = \frac{h\tan\alpha}{\tan\beta - \tan\alpha}$$

Hence, height of the building $= \frac{h\tan\alpha}{\tan\beta - \tan\alpha}$

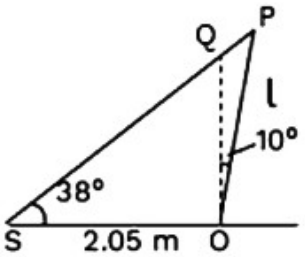
Q. **Length of shadow of pole is observed when elevation of light source is 38°. Length shadow of a pole is 2.05 meters when pole is inclined at 10° vertically towards light source. The length of the tower is**

(A) $\frac{2.05\sin 38^\circ}{\sin 42^\circ}$ **(B)** $\frac{2.05\sin 42^\circ}{\sin 38^\circ}$ **(C)** $\frac{2.05\cos 38^\circ}{\cos 42^\circ}$ **(D) None of these**

Sol **(A)**

$$\frac{\sin 38^\circ}{l} = \frac{\sin(\angle SPO)}{2.05}$$

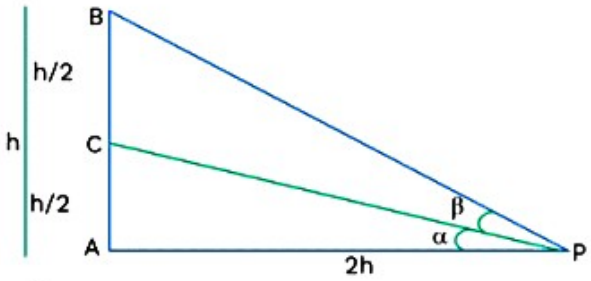
$$= \frac{\sin(180^\circ - 38^\circ - 90^\circ - 10^\circ)}{2.05} \Rightarrow l = \frac{2.05\sin 38^\circ}{\sin 42^\circ}$$



Q. **Let a vertical tower AB have its end A on the level ground. Let C be the mid point of AB and P be a point on the ground such that AP = 2AB. if $\angle BPC = \beta$, then $\tan\beta$ is equal to**

(A) $\frac{1}{4}$ **(B)** $\frac{6}{7}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{2}{9}$

Sol. In given figure



Let AB = h & AP = 2h

Let $\angle CPA = \alpha$

In ΔABP

$$\tan(\alpha+\beta) = \frac{h}{2h} = \frac{1}{2}$$

$$\frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{1}{2} \quad \text{...(i)}$$

In ΔAPC

$$\tan\alpha = \frac{AC}{AP} = \frac{h/2}{2h} = \frac{1}{4} \quad \text{...(ii)}$$

from (i) & (ii)

$$\frac{\frac{1}{4} + \tan\beta}{1 - \frac{\tan\beta}{4}} = \frac{1}{2}$$

$$\frac{1 + 4\tan\beta}{4 - \tan\beta} = \frac{1}{2}$$

$2 + 8\tan\beta = 4 - \tan\beta$

$9\tan\beta = 2$

$\Rightarrow \tan\beta = \frac{2}{9}$

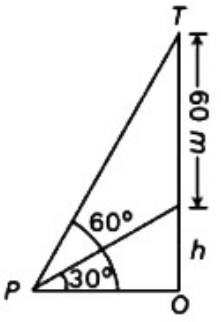
Q. **An observer in a boat finds that the angle of elevation of the top of cliff is 30° and that of a tower standing on the top of a cliff is 60°. If the height of the tower is 60 meters, then the height of the cliff is**

(A) 30 m **(B) $60\sqrt{3}$ m** **(C) $20\sqrt{3}$ m** **(D) None of these**

Sol **(A)**

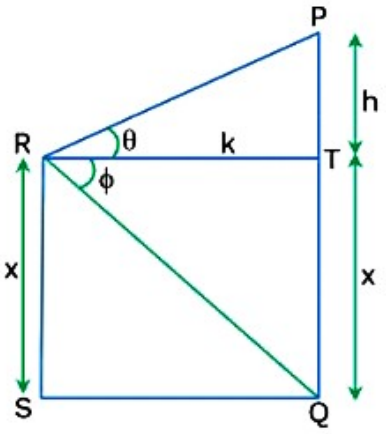
$(60 + h)\cot 60^\circ = h\cot 30^\circ$

$\Rightarrow\ h = 30m$



Q. **From a pole x meter above the ground angle of elevation & depression of top & foot of tower be θ & ϕ respectively. Show that height of tower is $x(1 + \tan\theta \cot\phi)$.**

Sol. Let k be the height of pole & x + h be the height of tower let distance between tower & pole be k.



In ΔRTQ

$$\tan\phi = \frac{QT}{RT} = \frac{x}{k}$$

$k = x\cot\phi$ ...(i)

Now In ΔPRT

$$\tan\theta = \frac{PT}{RT}$$

$$\tan\theta = \frac{h}{k} \Rightarrow k = h\cot\theta$$ ...(ii)

from (i) & (ii) $h = x\cot\phi\tan\theta$

So height t of tower is

$x + h = x + x\cot\phi\tan\theta$

$= x(1 + \cot\phi\tan\theta)$